

ITA0612 – MACHINE LEARNING

CO5-AT2

PSEUDOCODE ALGORITHM

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1. PSEUDOCODE – DATA CLEANING AND TRANSFORMATION ON TIME-SERIES DATA

Algorithm: Data_Cleaning_And_Transformation

BEGIN

 INPUT time_series_data

 CONVERT timestamp into standard date-time format

 REMOVE duplicate records

 SORT data by timestamp in ascending order

 FIND minimum and maximum timestamps

 CREATE complete time index at fixed interval

 REINDEX data using complete time index

 FOR each record in data DO

 IF data value is missing THEN

 INTERPOLATE value using neighboring values

 END IF

 END FOR

 FOR each remaining missing value DO

 IF previous valid value exists THEN

 FORWARD FILL the missing value

 ELSE

 BACKWARD FILL the missing value

 END IF

 END FOR

 FOR each data value DO

 IF value is invalid OR outside allowed range THEN

 MARK value as invalid

 END IF

 END FOR

 REMOVE invalid records

 RESAMPLE data at the required fixed interval

 APPLY appropriate aggregation function

 such as MEAN, SUM, MIN, or MAX

```
    SORT data by timestamp
    CHECK for remaining missing values
    CHECK for duplicate timestamps
    CHECK chronological order
    OUTPUT cleaned and transformed data
END
```

2. PSEUDOCODE – FEATURE SELECTION

Algorithm: Feature_Selection

```
BEGIN
    INPUT dataset
    SEPARATE features X and target variable Y
    HANDLE missing values in X
    NORMALIZE or STANDARDIZE numerical features
    CALCULATE correlation matrix for all features
    SET correlation_threshold = 0.80
    FOR each pair of features  $F_i$  and  $F_j$  DO
        IF ABSOLUTE(CORRELATION( $F_i$ ,  $F_j$ ))  $\geq$  correlation_threshold THEN
            IDENTIFY  $F_i$  and  $F_j$  as redundant features
            COMPARE correlation of  $F_i$  and  $F_j$  with target Y
            REMOVE the feature having lower correlation with Y
        END IF
    END FOR
    FOR each remaining feature  $F_i$  DO
        CALCULATE correlation between  $F_i$  and target Y

        IF ABSOLUTE(CORRELATION( $F_i$ , Y))  $<$  target_threshold THEN
            REMOVE  $F_i$ 
        ELSE
            RETAIN  $F_i$ 
        END IF
    END FOR
    STORE selected features
```

```
SPLIT dataset into training and validation sets
TRAIN machine learning model using selected features
PREDICT target values using validation data
CALCULATE model performance
IF model performance is improved THEN
    ACCEPT selected features
ELSE
    MODIFY selection threshold
    REPEAT feature selection process
END IF
OUTPUT selected features
END
```